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SET -7

13. Write a Pandas program to detect missing values of a given DataFrame. Display True or False.

AIM: To Write a Pandas program to detect missing values of a given DataFrame. Display True or False.

PROCEDURE:

- Import the Pandas and NumPy libraries.
- Create a DataFrame with some **NaN** values.
- Display the original DataFrame.
- Use the **isnull()** function to detect missing values.
- Display the result as **True** or **False**.

OUTPUT:

The screenshot shows a Jupyter Notebook interface with a file explorer on the left, a code editor in the center, and a terminal at the bottom. The code in the notebook is as follows:

```
experiment13.py > ...
10 })
11
12 print("Original DataFrame")
13 print(df)
14
15 # Detect missing values
16 print("\nMissing Values (True/False)")
17 print(df.isnull())
```

The terminal output shows the original DataFrame and the result of the `isnull()` function:

```
D:\Desktop\Query_Processing_Lab>python experiment13.py
Original DataFrame
   A    B    C    D
0  10.0  15.0  5.0  100.0
1  20.0  NaN  10.0  200.0
2  NaN  35.0  15.0  300.0
3  40.0  45.0  NaN  400.0
4  50.0  55.0  25.0   NaN

Missing Values (True/False)
   A    B    C    D
0  False False False False
1  False  True False False
2  True  False False False
3  False False  True False
4  False False False  True
```

The terminal also shows a notification to install the 'Rainbow CSV' extension from mechatroner for sales.csv.

RESULT: Thus, the Pandas program to detect missing values in a given DataFrame and display the results as True or False was executed successfully.

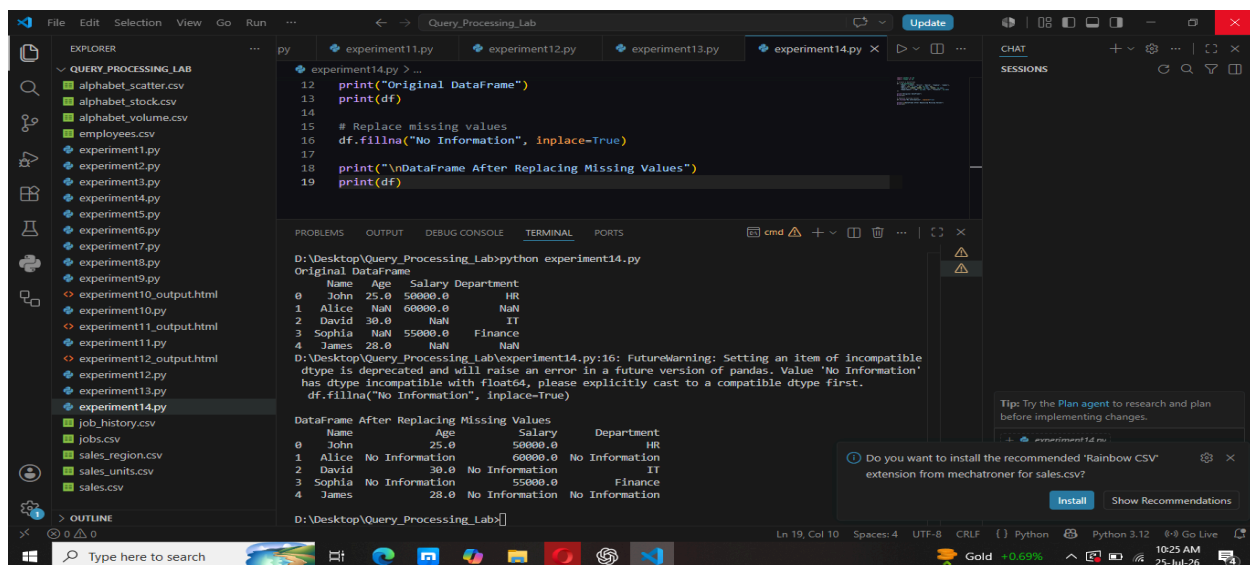
14. Write a Pandas program to find and replace the missing values in a given DataFrame which do not have any valuable information.

AIM: To Write a Pandas program to find and replace the missing values in a given DataFrame which do not have any valuable information.

PROCEDURE:

- Import the Pandas and NumPy libraries.
- Create a DataFrame containing some missing (NaN) values.
- Display the original DataFrame.
- Use the `fillna()` function to replace all missing values with "No Information".
- Display the updated DataFrame.

OUTPUT:



The screenshot shows a Jupyter Notebook interface with a file explorer on the left, a code editor in the center, and a terminal at the bottom. The code in the notebook is as follows:

```
12 print("Original DataFrame")
13 print(df)
14
15 # Replace missing values
16 df.fillna("No Information", inplace=True)
17
18 print("\nDataFrame After Replacing Missing Values")
19 print(df)
```

The terminal output shows the original DataFrame and the updated DataFrame after replacing missing values with "No Information".

Original DataFrame

	Name	Age	Salary	Department
0	John	25.0	50000.0	HR
1	Alice	NaN	60000.0	NaN
2	David	30.0	NaN	IT
3	Sophia	NaN	55000.0	Finance
4	James	28.0	NaN	NaN

DataFrame After Replacing Missing Values

	Name	Age	Salary	Department
0	John	25.0	50000.0	HR
1	Alice	No Information	60000.0	No Information
2	David	30.0	No Information	IT
3	Sophia	No Information	55000.0	Finance
4	James	28.0	No Information	No Information

RESULT: Thus, the Pandas program to find and replace the missing values in a given DataFrame with "No Information" was executed successfully, and the required output was obtained.